2.48 2nd Order DEs: Damped Motion

Worked Solutions

1) a)

The restoring force acts towards O, so it acts in the opposite direction to displacement:

$$\text{Restoring Force} = -10x \text{ N}.$$

1) b)

The resistance force always opposes motion, so if the velocity $v = \dfrac{dx}{dt}$ then

$$\text{Resistance Force} = -6v = -6\frac{dx}{dt} \text{ N}.$$

This makes sense as the resistance force is negative when moving to the right (as $\dfrac{dx}{dt}$ will be positive), and positive when moving to the left (as $\dfrac{dx}{dt}$ will be negative), and so it always opposes motion. Therefore, Resistance Force $= -6\dfrac{dx}{dt}$ N.

1) c)

$$\begin{aligned} F = ma, \qquad m = 2, \quad a = \frac{dv}{dt} = \frac{d^2x}{dt^2}, \\ -10x - 6\frac{dx}{dt} = 2\frac{d^2x}{dt^2}, \\ 2\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 10x = 0, \\ \frac{d^2x}{dt^2} + 3\frac{dx}{dt} + 5x = 0. \end{aligned}$$

2)

The positive direction is away from A, so the force of magnitude $4x$ N acting towards A will be $-4x$ N. The resistance force will act against the direction of motion, so $-2v = -2\dfrac{dx}{dt}$. Thus

$$\begin{aligned} F = ma, \qquad m = 0.5, \quad a = \frac{dv}{dt} = \frac{d^2x}{dt^2}, \\ -4x - 2\frac{dx}{dt} = 0.5\frac{d^2x}{dt^2}, \\ 0.5\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 4x = 0, \\ \frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 8x = 0. \end{aligned}$$

3)

The positive direction is away from O. There is a force acting on P directed towards O, so $F = -kx$ N. The resistance force acts against the direction of motion, so $-\lambda v = -\lambda\dfrac{dx}{dt}$ N. Since $a = \dfrac{dv}{dt} = \dfrac{d^2x}{dt^2}$,

$$-kx - \lambda\frac{dx}{dt} = m\frac{d^2x}{dt^2}, \qquad m\frac{d^2x}{dt^2} + \lambda\frac{dx}{dt} + kx = 0.$$

4)

The restoring force is $-12x$ N, the resistance force is $-9v = -9\dfrac{dx}{dt}$ N, and the driving force is $15\cos t$ N. With $m = 3$ and $a = \dfrac{d^2x}{dt^2}$,

$$\begin{aligned} -12x - 9\frac{dx}{dt} + 15\cos t &= 3\frac{d^2x}{dt^2}, \\ 3\frac{d^2x}{dt^2} + 9\frac{dx}{dt} + 12x &= 15\cos t, \\ \frac{d^2x}{dt^2} + 3\frac{dx}{dt} + 4x &= 5\cos t. \end{aligned}$$

5) a)

The restoring force is $-8x$ N and the resistance force is $-4v = -4\dfrac{dx}{dt}$ N. With $m = 1$,

$$-8x - 4\frac{dx}{dt} = \frac{d^2x}{dt^2} \quad \Longrightarrow \quad \frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 8x = 0.$$

5) b)

The auxiliary equation is

$$m^2 + 4m + 8 = 0 \Longrightarrow (m+2)^2 = -4 \Longrightarrow m = -2 \pm 2i.$$

Hence $x = e^{-2t}(A\cos 2t + B\sin 2t)$. Using $x = 2$ at $t = 0$ gives $A = 2$, so

$$x = e^{-2t}(2\cos 2t + B\sin 2t).$$

Differentiating,

$$\frac{dx}{dt} = e^{-2t}(-4\sin 2t + 2B\cos 2t) - 2e^{-2t}(2\cos 2t + B\sin 2t).$$

Using $\dfrac{dx}{dt} = 0$ at $t = 0$ gives $0 = 2B - 4$, so $B = 2$. Therefore

$$x = e^{-2t}(2\cos 2t + 2\sin 2t) = 2e^{-2t}(\cos 2t + \sin 2t).$$

6)

The restoring force is $-6x$ N and the resistive force is $-8v = -8\dfrac{dx}{dt}$ N. With $m = 2$,

$$\begin{aligned} -6x - 8\frac{dx}{dt} &= 2\frac{d^2x}{dt^2}, \\ \frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 3x &= 0. \end{aligned}$$

The auxiliary equation gives $(m+3)(m+1) = 0$, so the general solution is $x = Ae^{-3t} + Be^{-t}$. Using $x = 3$ at $t = 0$ gives $A + B = 3$. Also $\dfrac{dx}{dt} = -3Ae^{-3t} - Be^{-t}$, and using $\dfrac{dx}{dt} = -1$ at $t = 0$ gives $3A + B = 1$. Solving gives $A = -1$ and $B = 4$, hence

$$x = -e^{-3t} + 4e^{-t}.$$

7) a)

The force of magnitude $5x$ N acts in the direction PO, so it is $-5x$ N. The resistance force is $-2v = -2\dfrac{dx}{dt}$ N. With $m = 0.8$,

$$\begin{aligned} -5x - 2\frac{dx}{dt} &= 0.8\frac{d^2x}{dt^2}, \\ 0.8\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 5x &= 0, \\ 4\frac{d^2x}{dt^2} + 10\frac{dx}{dt} + 25x &= 0. \end{aligned}$$

7) b)

The auxiliary equation is

$$\begin{aligned} 4m^2 + 10m + 25 &= 0, \\ 4\left(m + \frac{5}{4}\right)^2 &= -\frac{75}{4}, \\ m &= -\frac{5}{4} \pm \frac{5\sqrt{3}}{4}i. \end{aligned}$$

Thus $x = e^{-5t/4}\left(A\cos\frac{5\sqrt{3}}{4}t + B\sin\frac{5\sqrt{3}}{4}t\right)$. Using $x = 1.2$ at $t = 0$ gives $A = 1.2$. Differentiating and using $\dfrac{dx}{dt} = 3$ at $t = 0$ gives $B = \dfrac{6\sqrt{3}}{5}$. Therefore

$$x = e^{-5t/4}\left(\frac{6}{5}\cos\frac{5\sqrt{3}}{4}t + \frac{6\sqrt{3}}{5}\sin\frac{5\sqrt{3}}{4}t\right), \qquad x\big|_{t=1} = 0.301 \text{ m (3 SF)}.$$

8) a)

The restoring force is $-7.5x$ N, the resistance force is $-6\dfrac{dx}{dt}$ N, and the driving force is $9\sin 2t$ N. With $m = 1.5$,

$$-7.5x - 6\frac{dx}{dt} + 9\sin 2t = 1.5\frac{d^2x}{dt^2} \quad \Longrightarrow \quad \frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 5x = 6\sin 2t.$$

8) b)

The homogeneous equation has auxiliary equation $m^2 + 4m + 5 = 0$, giving $m = -2 \pm i$. Thus the complementary function is $x = e^{-2t}(A\cos t + B\sin t)$. Try $x = p\cos 2t + q\sin 2t$ for a particular integral. Then

$$\begin{aligned} x' &= -2p\sin 2t + 2q\cos 2t, \\ x'' &= -4p\cos 2t - 4q\sin 2t, \\ x'' + 4x' + 5x &= (p + 8q)\cos 2t + (-8p + q)\sin 2t. \end{aligned}$$

Comparing coefficients gives $p + 8q = 0$ and $-8p + q = 6$, so $p = -\dfrac{48}{65}$ and $q = \dfrac{6}{65}$. Hence the general solution is

$$x = e^{-2t}(A\cos t + B\sin t) - \frac{48}{65}\cos 2t + \frac{6}{65}\sin 2t.$$

Using $x = 4$ at $t = 0$ gives $A = \dfrac{308}{65}$. Differentiating and using $x' = 1$ at $t = 0$ gives $B = \dfrac{669}{65}$. Therefore

$$x = e^{-2t}\left(\frac{308}{65}\cos t + \frac{669}{65}\sin t\right) - \frac{48}{65}\cos 2t + \frac{6}{65}\sin 2t.$$

8) c)

As $t \to \infty$, $e^{-2t} \to 0$, so $x \to -\frac{48}{65}\cos 2t + \frac{6}{65}\sin 2t$. The motion approaches steady oscillation (simple harmonic motion).

9) a)

The auxiliary equation is $m^2 + 6m + 25 = 0$, so $m = -3 \pm 4i$ and

$$x = e^{-3t}(A\cos 4t + B\sin 4t).$$

Taking downwards as positive and using $x = 0.08$ at $t = 0$ gives $A = 0.08$. Differentiating and using $x' = 0$ at $t = 0$ gives $B = 0.06$. Therefore

$$x = e^{-3t}(0.08\cos 4t + 0.06\sin 4t).$$

9) b)

The roots of the auxiliary equation are $-3 \pm 4i$, so the solution contains sine and cosine terms. Therefore the system undergoes damped oscillations (the real parts of the roots are negative).

9) c)

$$x' = e^{-3t}(-0.32\sin 4t + 0.24\cos 4t) - 3e^{-3t}(0.08\cos 4t + 0.06\sin 4t).$$

At a stationary point $x' = 0$, and therefore

$$\begin{aligned} 0 &= -0.32\sin 4t + 0.24\cos 4t - 3(0.08\cos 4t + 0.06\sin 4t), \\ 0 &= -0.5\sin 4t, \qquad \sin 4t = 0. \end{aligned}$$

The first positive time is $4t = \pi$, so $t = \frac{\pi}{4}$ seconds.

10) a)

The auxiliary equation is $m^2 + 2m + 10 = 0$, so $m = -1 \pm 3i$. The complementary function is $\theta = e^{-t}(A\cos 3t + B\sin 3t)$. Using $\theta = 0.2$ at $t = 0$ gives $A = 0.2$. Differentiating and using $\theta' = 0$ at $t = 0$ gives $B = \frac{1}{15}$. Hence

$$\theta = e^{-t}\left(0.2\cos 3t + \frac{1}{15}\sin 3t\right).$$

10) b)

As the auxiliary equation has two complex conjugate roots, the system undergoes light damping (underdamping).

10) c)

As $t \to \infty$, $e^{-t} \to 0$ so $\theta \to 0$. The panel oscillates with decreasing amplitude and tends towards rest at equilibrium.

11) a)

The restoring force is $-18x$ N and the resistance force is $-\lambda v = -\lambda\frac{dx}{dt}$ N. Thus

$$-18x - \lambda\frac{dx}{dt} = m\frac{d^2x}{dt^2} \quad \Longrightarrow \quad m\frac{d^2x}{dt^2} + \lambda\frac{dx}{dt} + 18x = 0.$$

11) b)

The auxiliary equation is $mr^2 + \lambda r + 18 = 0$.

11) c(i))

When $m = 2$ and $\lambda = 8$,

$$2r^2 + 8r + 18 = 0 \Longrightarrow (r+2)^2 = -5 \Longrightarrow r = -2 \pm \sqrt{5}i.$$

Thus $x = e^{-2t}(A\cos(\sqrt{5}t) + B\sin(\sqrt{5}t))$.

11) c(ii))

The roots are $-2 \pm \sqrt{5}i$, so the solution contains sine and cosine terms. Therefore the system undergoes damped oscillations (the real parts of the roots are negative).

11) d)

For $m = 2$, the auxiliary equation is $2r^2 + \lambda r + 18 = 0$. Critical damping requires $b^2 - 4ac = 0$, so $\lambda^2 - 4 \times 2 \times 18 = 0$, giving $\lambda = \pm 12$. Since $\lambda > 0$, $\lambda = 12$.

11) e)

If $m = 3$ and $\lambda = 12$,

$$3r^2 + 12r + 18 = 0 \Longrightarrow (r+2)^2 = -2 \Longrightarrow r = -2 \pm \sqrt{2}i.$$

As the roots are complex and their real parts are negative, the motion is light damping (underdamping), where the system undergoes damped oscillations.

12)

The differential equation is $2x'' + 7kx' + 6k^2x = 4kU$. The homogeneous equation has auxiliary equation

$$2m^2 + 7km + 6k^2 = 0 \Longrightarrow \left(m + \frac{7k}{4}\right)^2 = \frac{k^2}{16},$$

so $m = -\dfrac{3k}{2}$ or $m = -2k$. The complementary function is $x = Ae^{-3kt/2} + Be^{-2kt}$. For a constant particular integral $x = c$, $6k^2c = 4kU$, so $c = \dfrac{2U}{3k}$. Hence

$$x = Ae^{-3kt/2} + Be^{-2kt} + \frac{2U}{3k}.$$

Using $x = 0$ at $t = 0$ gives $A + B = -\dfrac{2U}{3k}$. Also

$$x' = -\frac{3}{2}Ake^{-3kt/2} - 2Bke^{-2kt},$$

and using $x' = 0$ at $t = 0$ gives $\dfrac{3}{2}A + 2B = 0$, so $A = -\dfrac{4}{3}B$. Substitution gives $B = \dfrac{2U}{k}$ and $A = -\dfrac{8U}{3k}$. Therefore

$$x = -\frac{8U}{3k}e^{-3kt/2} + \frac{2U}{k}e^{-2kt} + \frac{2U}{3k}.$$

13) a)

Taking downwards as positive, as in the printed solution, the auxiliary equation for the homogeneous equation is $m^2 + 4m + 4 = 0$, so $(m+2)^2 = 0$ and the complementary function is $x = Ate^{-2t} + Be^{-2t}$. Try $x = pt^2e^{-2t}$ for a particular integral. Then

$$x' = -2pt^2e^{-2t} + 2pte^{-2t},$$
$$x'' = 4pt^2e^{-2t} - 8pte^{-2t} + 2pe^{-2t}.$$

Substituting into $x'' + 4x' + 4x = 10e^{-2t}$ gives $2pe^{-2t} = 10e^{-2t}$, so $p = 5$. Thus the general solution is $x = Ate^{-2t} + Be^{-2t} + 5t^2e^{-2t}$. Using $x = 18$ at $t = 0$ gives $B = 18$. Differentiating and using $x' = 8$ at $t = 0$ gives $A = 44$. Therefore

$$x = (5t^2 + 44t + 18)e^{-2t}.$$

13) b)

$$x' = (10t + 44)e^{-2t} - 2(5t^2 + 44t + 18)e^{-2t}.$$

At a maximum, $x' = 0$, so

$$\begin{aligned}(10t + 44) - 2(5t^2 + 44t + 18) &= 0,\\ 5t^2 + 39t - 4 &= 0,\\ t &= \frac{-39 \pm \sqrt{1601}}{10} = -7.901249\ldots \text{ or } 0.101249\ldots.\end{aligned}$$

Since $t \geq 0$, use $t = 0.101249\ldots$. Then $x = 18.380556\ldots = 18.4$ cm to the nearest mm.

13) c)

The mass is at equilibrium when $x = 0$. Since $e^{-2t} \neq 0$ for all $t \geq 0$, this would require $5t^2+44t+18 = 0$. However, for $t \geq 0$, $5t^2 \geq 0$, $44t \geq 0$ and $18 > 0$, so $5t^2 + 44t + 18 > 0$. Hence $x > 0$ for all $t \geq 0$, and the model predicts that the mass never returns to its equilibrium position.

14) a)

The homogeneous equation has auxiliary equation $m^2 + 6m + 13 = 0$, giving $m = -3 \pm 2i$. The complementary function is $x = e^{-3t}(A\cos 2t + B\sin 2t)$. Try $x = p\cos 2t + q\sin 2t$ for a particular integral. Then

$$\begin{aligned}x' &= -2p\sin 2t + 2q\cos 2t,\\ x'' &= -4p\cos 2t - 4q\sin 2t,\\ x'' + 6x' + 13x &= (9p + 12q)\cos 2t + (-12p + 9q)\sin 2t.\end{aligned}$$

Comparing coefficients gives $9p + 12q = 10$ and $-12p + 9q = 0$, so $p = \frac{2}{5}$ and $q = \frac{8}{15}$. Therefore

$$x = e^{-3t}(A\cos 2t + B\sin 2t) + \frac{2}{5}\cos 2t + \frac{8}{15}\sin 2t.$$

14) b)

The steady-state solution is the part that remains as t becomes large, so

$$x = \frac{2}{5}\cos 2t + \frac{8}{15}\sin 2t.$$

14) c)

The amplitude is

$$\sqrt{\left(\frac{2}{5}\right)^2 + \left(\frac{8}{15}\right)^2} = \frac{2}{3} \text{ m}.$$

15) a)

The homogeneous equation has auxiliary equation $m^2 + 8m + 32 = 0$, giving $m = -4 \pm 4i$. The complementary function is $\theta = e^{-4t}(A\cos 4t + B\sin 4t)$. Try $\theta = p\cos 4t + q\sin 4t$ for a particular integral. Then

$$\begin{aligned}\theta' &= -4p\sin 4t + 4q\cos 4t,\\ \theta'' &= -16p\cos 4t - 16q\sin 4t,\\ \theta'' + 8\theta' + 32\theta &= (16p + 32q)\cos 4t + (-32p + 16q)\sin 4t.\end{aligned}$$

Comparing coefficients gives $16p + 32q = 80$ and $-32p + 16q = 0$, so $p = 1$ and $q = 2$. The general solution is

$$\theta = e^{-4t}(A\cos 4t + B\sin 4t) + \cos 4t + 2\sin 4t.$$

15) b)

Using $\theta = 0$ at $t = 0$ gives $A = -1$. Differentiating and using $\theta' = 9$ at $t = 0$ gives $B = -\frac{3}{4}$. Hence

$$\theta = e^{-4t}\left(-\cos 4t - \frac{3}{4}\sin 4t\right) + \cos 4t + 2\sin 4t.$$

15) c)

As $t \to \infty$, $e^{-4t} \to 0$, so $\theta \to \cos 4t + 2\sin 4t$. As t becomes large, the pendulum approaches simple harmonic motion.

16)

The auxiliary equation is $m^2 + 6m + 9 = 0$, so $(m + 3)^2 = 0$. The complementary function is $\theta = Ate^{-3t} + Be^{-3t}$. Try $\theta = pe^{-t}$ for a particular integral. Substitution gives $4pe^{-t} = 9e^{-t}$, so $p = \frac{9}{4}$. Hence

$$\theta = Ate^{-3t} + Be^{-3t} + \frac{9}{4}e^{-t}.$$

Using $\theta = 0$ at $t = 0$ gives $B = -\frac{9}{4}$. Differentiating and using $\theta' = 0$ at $t = 0$ gives $A = -\frac{9}{2}$. Therefore

$$\theta = -\frac{9}{2}te^{-3t} - \frac{9}{4}e^{-3t} + \frac{9}{4}e^{-t}.$$

The pendulum is at the downward vertical when $\theta = 0$. Thus

$$\begin{aligned}0 &= \left(-\frac{1}{2}t - \frac{9}{4} + \frac{9}{4}e^{2t}\right)e^{-3t},\\ 0 &= 2t + 1 - e^{2t}.\end{aligned}$$

Since $e^{-3t} \neq 0$, this would require $e^{2t} - 2t - 1 = 0$. The tangent to $y = e^{2t}$ at $(0, 1)$ is $y = 2t + 1$. Since $e^{2t} > 2t + 1$ for all $t > 0$, the equation has no solution for $t > 0$. Hence $\theta > 0$ for all $t > 0$, and the pendulum never returns to the downward vertical.

17) a)

Take the positive direction to be the direction in which the spring is being compressed. The spring force is $-M\alpha^2 y$ N, so

$$-M\alpha^2 y = M\frac{d^2y}{dt^2} \quad \Longrightarrow \quad \frac{d^2y}{dt^2} + \alpha^2 y = 0.$$

17) b)

At the instant the door first touches the spring, it has not yet been compressed, so $y = 0$ when $t = 0$. The door moves into the spring with speed W, and the direction of compression is positive, so $\left.\dfrac{dy}{dt}\right|_{t=0} = W$.

17) c)

The auxiliary equation is $m^2 + \alpha^2 = 0$, so $m = \pm\alpha i$ and $y = A\cos\alpha t + B\sin\alpha t$. Using $y = 0$ at $t = 0$ gives $A = 0$. Then $y = B\sin\alpha t$ and $y' = B\alpha\cos\alpha t$. Using $y' = W$ at $t = 0$ gives $B = \dfrac{W}{\alpha}$. Therefore $y = \dfrac{W}{\alpha}\sin\alpha t$.

17) d)

Since the maximum value of $\sin\alpha t$ is 1, the maximum compression is $\dfrac{W}{\alpha}$.

17) e)

The spring force is $-M\alpha^2 y$ N and the resistive force is $-\lambda Mv = -\lambda M\dfrac{dy}{dt}$. Hence

$$-M\alpha^2 y - \lambda M\frac{dy}{dt} = M\frac{d^2y}{dt^2} \quad \Longrightarrow \quad \frac{d^2y}{dt^2} + \lambda\frac{dy}{dt} + \alpha^2 y = 0.$$

17) f)

When $\lambda = 3\alpha$, the auxiliary equation is $m^2 + 3\alpha m + \alpha^2 = 0$. Its discriminant is

$$(3\alpha)^2 - 4 \times 1 \times \alpha^2 = 5\alpha^2 > 0,$$

since α is a positive real constant. Thus the auxiliary equation has two distinct real roots, and the motion is heavy damping (overdamped).

17) g)

The door compresses the spring and is slowed by damping. It does not oscillate, but approaches equilibrium without repeated overshoot.

18) a)

The homogeneous equation has auxiliary equation $m^2 + 4m + 13 = 0$, so $m = -2 \pm 3i$ and $\theta = e^{-2t}(A\cos 3t + B\sin 3t)$. A constant particular integral $\theta = k$ gives $13k = 26$, so $k = 2$. Using $\theta = 3$ at $t = 0$ gives $A = 1$. Differentiating and using $\theta' = 0$ at $t = 0$ gives $B = \dfrac{2}{3}$. Thus

$$\theta = e^{-2t}\left(\cos 3t + \frac{2}{3}\sin 3t\right) + 2.$$

18) b)

$$\cos 3t + \frac{2}{3}\sin 3t \equiv R\cos(3t - \alpha) = R\cos\alpha\cos 3t + R\sin\alpha\sin 3t.$$

Thus $R\cos\alpha = 1$ and $R\sin\alpha = \dfrac{2}{3}$, so

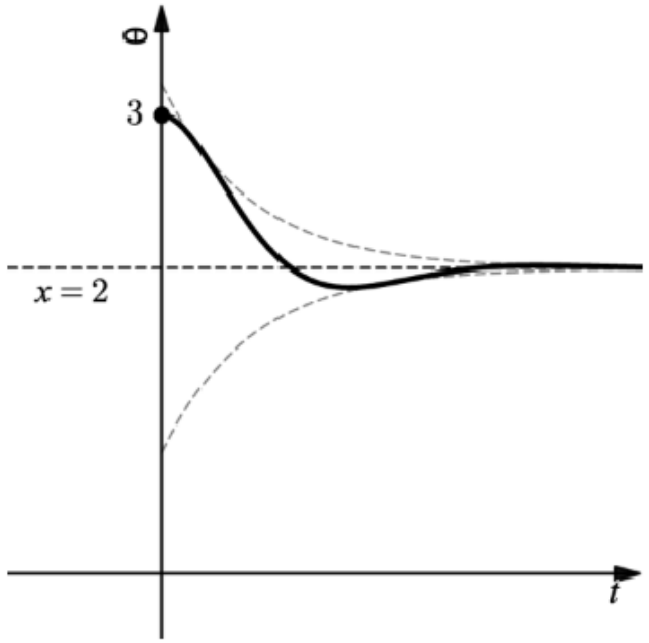
$$R = \sqrt{1^2 + \left(\frac{2}{3}\right)^2} = \frac{\sqrt{13}}{3}, \qquad \alpha = \arctan\frac{2}{3} = 0.588 \text{ (3 SF)}.$$

Therefore $\theta = 2 + \dfrac{\sqrt{13}}{3}e^{-2t}\cos\left(3t - \arctan\dfrac{2}{3}\right)$.

18) c)

As $t \to \infty$, $e^{-2t} \to 0$ and $\theta \to 2$. The oscillations die away because of the damping, and the door settles to a constant angle of 2 radians from the vertical.

18) d)



19) a)

Take downwards as positive. The weight is Mg N, the tension is $-Cy$ N, and the resistive force is $-Dv = -D\dfrac{dy}{dt}$. Thus

$$Mg - Cy - D\frac{dy}{dt} = M\frac{d^2y}{dt^2} \quad \Longrightarrow \quad M\frac{d^2y}{dt^2} + D\frac{dy}{dt} + Cy = Mg.$$

19) b)

The homogeneous equation has auxiliary equation $Mr^2 + Dr + C = 0$. Completing the square gives

$$r = -\frac{D}{2M} \pm i\frac{\sqrt{4MC - D^2}}{2M},$$

since $4CM > D^2$. Hence the complementary function is

$$y = e^{-Dt/(2M)}\left(P\cos\left(\frac{\sqrt{4CM - D^2}}{2M}t\right) + Q\sin\left(\frac{\sqrt{4CM - D^2}}{2M}t\right)\right).$$

For a constant particular integral $y = k$, $Ck = Mg$, so $k = \dfrac{Mg}{C}$. Therefore

$$y = e^{-Dt/(2M)}\left(P\cos\left(\frac{\sqrt{4CM - D^2}}{2M}t\right) + Q\sin\left(\frac{\sqrt{4CM - D^2}}{2M}t\right)\right) + \frac{Mg}{C}.$$

19) c)

Substituting $C = 25$, $D = 24$, $M = 9$ and $g = 10$ gives

$$y = e^{-4t/3}(P\cos t + Q\sin t) + \frac{18}{5}.$$

Using $y = 0$ at $t = 0$ gives $P = -\frac{18}{5}$. Differentiating and using $y' = 9$ at $t = 0$ gives $Q = \frac{21}{5}$. Hence

$$y = e^{-4t/3}\left(-\frac{18}{5}\cos t + \frac{21}{5}\sin t\right) + \frac{18}{5}.$$

19) d)

The crate reaches its greatest extension when $y' = 0$. Differentiating the expression in part (c) and setting the derivative equal to zero gives

$$\begin{aligned} 0 &= \left(\frac{18}{5}\sin t + \frac{21}{5}\cos t\right) - \frac{4}{3}\left(-\frac{18}{5}\cos t + \frac{21}{5}\sin t\right), \\ 0 &= 9\cos t - 2\sin t, \\ \tan t &= \frac{9}{2}. \end{aligned}$$

Thus $t = \arctan\frac{9}{2} = 1.35$ s (3 SF).

19) e)

As $t \to \infty$, $e^{-4t/3} \to 0$, so $y \to \frac{18}{5}$. This is the equilibrium extension of the cable, 3.6 m.

19) f)

With no resistive force, $Mg - Cy = My''$, so $My'' + Cy = Mg$. Substituting $C = 25$, $M = 9$ and $g = 10$ gives $9y'' + 25y = 90$. The complementary function is $y = P\cos\frac{5}{3}t + Q\sin\frac{5}{3}t$, and the constant particular integral is $y = \frac{18}{5}$. Hence

$$y = P\cos\frac{5}{3}t + Q\sin\frac{5}{3}t + \frac{18}{5}.$$

Using $y = 0$ at $t = 0$ gives $P = -\frac{18}{5}$. Differentiating and using $y' = 9$ at $t = 0$ gives $Q = \frac{27}{5}$. Therefore

$$y = -\frac{18}{5}\cos\frac{5}{3}t + \frac{27}{5}\sin\frac{5}{3}t + \frac{18}{5}.$$

19) g)

For the damped model, $\tan t = \frac{9}{2}$, so $t = \arctan\frac{9}{2} = 1.352\ldots$. Substitution into the damped expression gives the maximum extension $4.147078965\ldots = 4.15$ m (3 SF). For the undamped model,

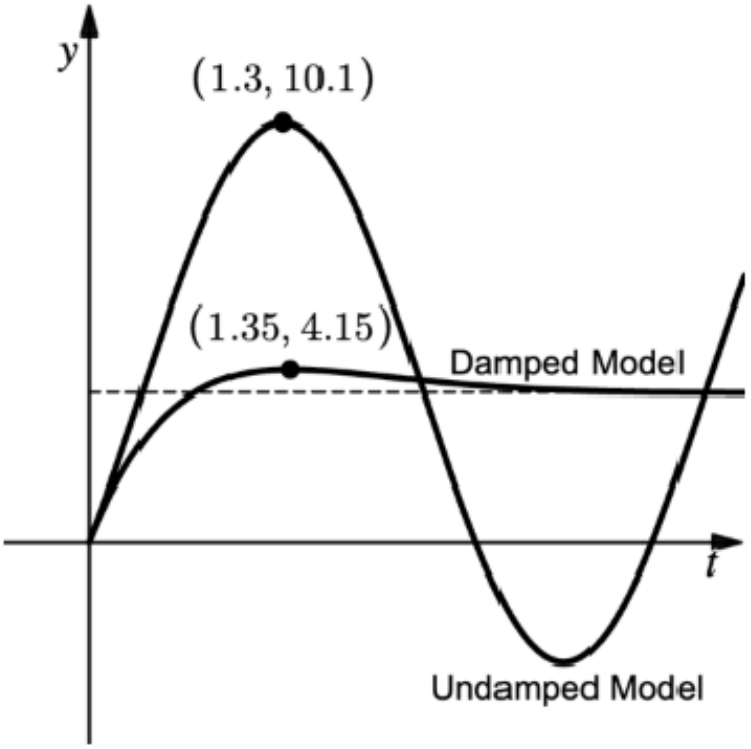
$$y = -\frac{18}{5}\cos\frac{5}{3}t + \frac{27}{5}\sin\frac{5}{3}t + \frac{18}{5}, \qquad y' = 6\sin\frac{5}{3}t + 9\cos\frac{5}{3}t.$$

The maximum occurs when $y' = 0$, so $\tan\frac{5}{3}t = -\frac{3}{2}$. The first positive time is

$$t = \frac{3}{5}\left(\arctan\left(-\frac{3}{2}\right) + \pi\right) = 1.295\ldots.$$

Substitution gives the maximum extension $10.0899923\ldots = 10.1$ m (3 SF). Since $10.1 > 4.15$, the greatest extension predicted by the undamped model is larger than that predicted by the damped model.

19) h)



20) a)

The equation of motion is $\ddot{x} + \lambda\dot{x} = 0$ and $v = \dot{x} = \dfrac{dx}{dt}$. Therefore

$$\ddot{x} = \frac{dv}{dt} = \frac{dv}{dx}\frac{dx}{dt} = v\frac{dv}{dx},$$

so $v\dfrac{dv}{dx} + \lambda v = 0$.

20) b)

Since the particle is moving, $v \neq 0$, so divide through by v:

$$\frac{dv}{dx} + \lambda = 0 \quad \Longrightarrow \quad v = -\lambda x + c.$$

Initially $x = 0$ and $v = u$, so $c = u$. Hence $v = u - \lambda x$.

20) c)

At A, the particle comes to instantaneous rest, so $v = 0$ when $x = d$. Thus $0 = u - \lambda d$, giving $u = \lambda d$.

20) d)

Using $u = \lambda d$ in $v = u - \lambda x$ gives $v = \lambda d - \lambda x = \lambda(d - x)$.

20) e)

Since $z = \dfrac{1}{v}$, we have $z = \dfrac{1}{\lambda(d-x)}$. The mean value of z with respect to displacement from $x = 0$ to $x = \dfrac{1}{2}d$ is

$$\begin{aligned}\frac{1}{\frac{1}{2}d - 0}\int_0^{d/2} \frac{1}{\lambda(d-x)}\,dx &= \frac{2}{\lambda d}\left[-\ln|d-x|\right]_0^{d/2} \\ &= \frac{2}{\lambda d}\ln\left(\frac{d}{d/2}\right) = \frac{2\ln 2}{\lambda d} = \frac{\ln 4}{u}.\end{aligned}$$

21) a)

The amplitude at time t is the coefficient of $\cos 3t$, so it is $5e^{-0.2t}$.

21) b)

The period is $T = \dfrac{2\pi}{3}$. Thus the ratio of the amplitudes at times t and $t+T$ is

$$\frac{5e^{-0.2(t+T)}}{5e^{-0.2t}} = e^{-0.2T} = e^{-2\pi/15}.$$

22) a)

The auxiliary equation is $m^2 + 2cm + 10 = 0$, so

$$m = -c \pm \sqrt{c^2 - 10}.$$

For oscillatory motion, $c^2 < 10$, and therefore $m = -c \pm i\sqrt{10 - c^2}$. The general solution is

$$y = e^{-ct}(A\cos\omega t + B\sin\omega t), \qquad \omega = \sqrt{10 - c^2}.$$

22) b)

One oscillation takes time $T = \dfrac{2\pi}{\omega} = \dfrac{2\pi}{\sqrt{10 - c^2}}$. After one oscillation the amplitude is multiplied by e^{-cT}. Since this is 0.9, $e^{-cT} = 0.9$.

22) c)

Using $T = \dfrac{2\pi}{\sqrt{10 - c^2}}$,

$$\begin{aligned} e^{-c(2\pi/\sqrt{10-c^2})} &= 0.9, \\ \frac{4\pi^2 c^2}{10 - c^2} &= (\ln 0.9)^2, \\ c^2 &= \frac{10(\ln 0.9)^2}{4\pi^2 + (\ln 0.9)^2}, \\ c &= 0.0530196636\ldots = 0.0530 \text{ (3 SF)}. \end{aligned}$$

22) d)

Using $y = 0$ at $t = 0$ gives $A = 0$, so $y = Be^{-ct}\sin\omega t$. Differentiating and using $y' = 15$ at $t = 0$ gives $B = \dfrac{15}{\omega}$. With $c = 0.0530196636\ldots$, $\omega = \sqrt{10 - c^2} = 3.16183315\ldots$ and $B = 4.74408333\ldots$. To three significant figures,

$$y = 4.74e^{-0.0530t}\sin 3.16t.$$

23) a)

Take the direction to the right as positive. The restoring force is $-4mx$ N, so

$$-4mx = m\frac{d^2x}{dt^2} \quad \Longrightarrow \quad \frac{d^2x}{dt^2} + 4x = 0.$$

23) b)

The auxiliary equation is $m^2 + 4 = 0$, so $m = \pm 2i$ and $x = A\cos 2t + B\sin 2t$. Using $x = 1$ at $t = 0$ gives $A = 1$. Differentiating and using $x' = 2$ at $t = 0$ gives $B = 1$. Thus $x = \cos 2t + \sin 2t$.

23) c)

The amplitude is $\sqrt{1^2+1^2}=\sqrt{2}$ and the period is $\dfrac{2\pi}{2}=\pi$.

23) d)

The motion is simple harmonic motion.

23) e)

The resistive force is $-2mv=-2m\dfrac{dx}{dt}$. Hence

$$-4mx-2m\frac{dx}{dt}=m\frac{d^2x}{dt^2} \quad\Longrightarrow\quad \frac{d^2x}{dt^2}+2\frac{dx}{dt}+4x=0.$$

23) f)

The auxiliary equation is $m^2+2m+4=0$, so $(m+1)^2=-3$ and $m=-1\pm\sqrt{3}i$. The general solution is $x=e^{-t}(A\cos\sqrt{3}t+B\sin\sqrt{3}t)$.

23) g)

The roots of the auxiliary equation are complex, so oscillations occur, but they decay with time. This is light damping (underdamping).

23) h)

Try $x=p\cos t+q\sin t$ for a particular integral. Then $x'=-p\sin t+q\cos t$ and $x''=-p\cos t-q\sin t$. Substitution into $x''+2x'+4x=4\cos t$ gives

$$(3p+2q)\cos t+(-2p+3q)\sin t=4\cos t.$$

Thus $3p+2q=4$ and $-2p+3q=0$, so $p=\dfrac{12}{13}$ and $q=\dfrac{8}{13}$. The general solution is

$$x=e^{-t}(A\cos\sqrt{3}t+B\sin\sqrt{3}t)+\frac{12}{13}\cos t+\frac{8}{13}\sin t.$$

Using $x=1$ at $t=0$ gives $A=\dfrac{1}{13}$. Differentiating and using $x'=2$ at $t=0$ gives $B=\dfrac{19}{13\sqrt{3}}=\dfrac{19\sqrt{3}}{39}$. Hence

$$x=e^{-t}\left(\frac{1}{13}\cos\sqrt{3}t+\frac{19\sqrt{3}}{39}\sin\sqrt{3}t\right)+\frac{12}{13}\cos t+\frac{8}{13}\sin t.$$

23) i)

As $t\to\infty$, $e^{-t}\to 0$, so $x\to\dfrac{12}{13}\cos t+\dfrac{8}{13}\sin t$. Its period is $2\pi=6.28318\ldots$ seconds, just over 6 seconds.

23) j)

The amplitude of the long-term motion is

$$\sqrt{\left(\frac{12}{13}\right)^2+\left(\frac{8}{13}\right)^2}=\frac{4\sqrt{13}}{13}.$$

23) k)

$$\frac{1}{13}\cos\sqrt{3}t + \frac{19\sqrt{3}}{39}\sin\sqrt{3}t \equiv C\cos(\sqrt{3}t - \beta).$$

Comparing coefficients gives $C\cos\beta = \dfrac{1}{13}$ and $C\sin\beta = \dfrac{19\sqrt{3}}{39}$. Therefore

$$C^2 = \frac{1}{169} + \frac{361}{507} = \frac{4\cdot 273}{39^2}, \qquad C = \frac{2\sqrt{273}}{39} > 0,$$

and $\tan\beta = \dfrac{19\sqrt{3}}{3}$, so $\beta = \arctan\left(\dfrac{19\sqrt{3}}{3}\right) = 1.48$ (3 SF). Thus

$$\frac{1}{13}\cos\sqrt{3}t + \frac{19\sqrt{3}}{39}\sin\sqrt{3}t \equiv \frac{2\sqrt{273}}{39}\cos\left(\sqrt{3}t - \arctan\frac{19\sqrt{3}}{3}\right).$$

23) l)

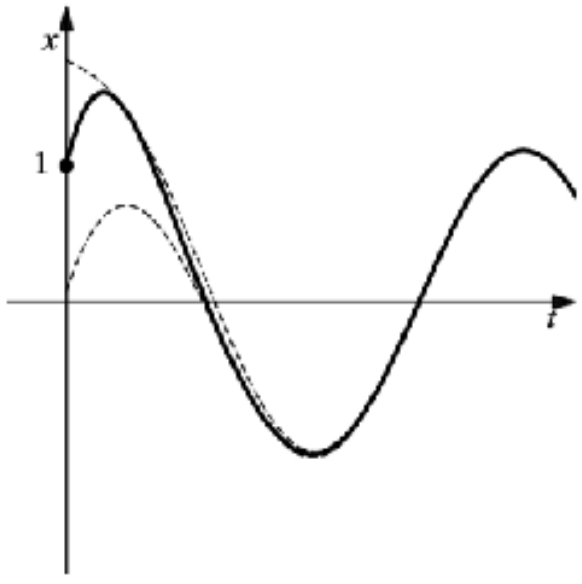
From part (k),

$$x(t) = e^{-t}\left(\frac{2\sqrt{273}}{39}\cos\left(\sqrt{3}t - \arctan\frac{19\sqrt{3}}{3}\right)\right) + x_s(t), \qquad x_s(t) = \frac{12}{13}\cos t + \frac{8}{13}\sin t.$$

Since $-1 \le \cos\left(\sqrt{3}t - \arctan\frac{19\sqrt{3}}{3}\right) \le 1$,

$$x_s(t) - \frac{2\sqrt{273}}{39}e^{-t} \le x(t) \le x_s(t) + \frac{2\sqrt{273}}{39}e^{-t}.$$

23) m)



24) a)

The complementary part $e^{-t/2}(A\cos 2t + B\sin 2t)$ corresponds to roots $m = -\dfrac{1}{2} \pm 2i$. Thus

$$\left(m + \frac{1}{2}\right)^2 = -4 \quad \Longrightarrow \quad m^2 + m + \frac{17}{4} = 0,$$

so it satisfies $x'' + x' + \frac{17}{4}x = 0$. For the remaining part $x = \cos t - \sin t$,

$$x' = -\sin t - \cos t, \qquad x'' = -\cos t + \sin t.$$

Then

$$\begin{aligned} x'' + x' + \frac{17}{4}x &= (-\cos t + \sin t) + (-\sin t - \cos t) + \frac{17}{4}(\cos t - \sin t) \\ &= \frac{9}{4}\cos t - \frac{17}{4}\sin t. \end{aligned}$$

Therefore

$$4\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 17x = 9\cos t - 17\sin t.$$

24) b)

